

Multi Layer Perceptron

Problem : What is the issue with perceptron?

The main issue with perceptron is that it can only learn linearly separable patterns.

It works by finding a linear boundary (decision boundary) that separates data points of different classes. So, if the data is NOT LINEAR, then a perceptron won't be able to correctly classify the data points.

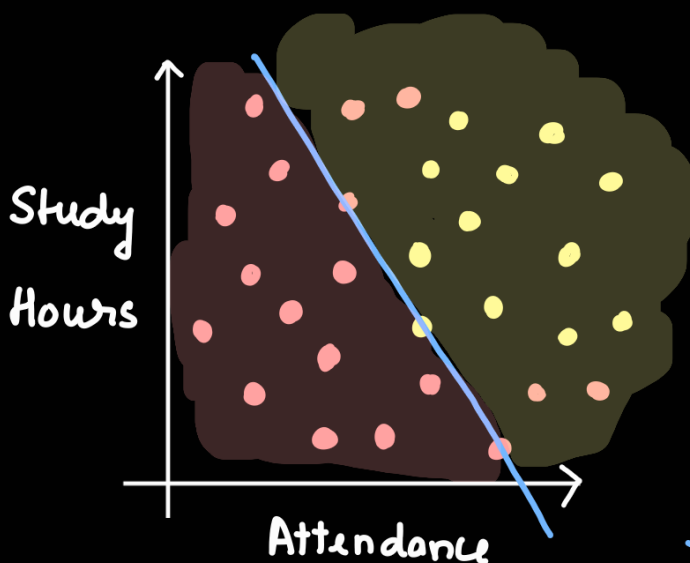
We have a STUDENT PERFORMANCE DATASET

FEATURES → Study Hours , Attendance

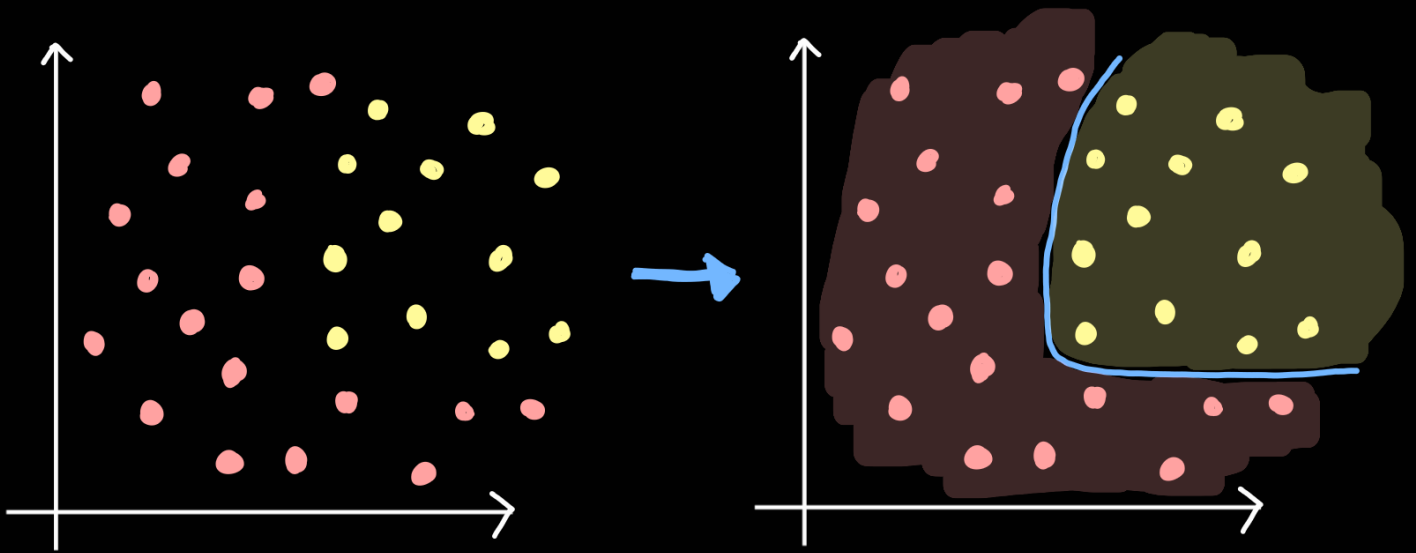
LABEL → Exam Pass Status , 0 for fail
1 for pass



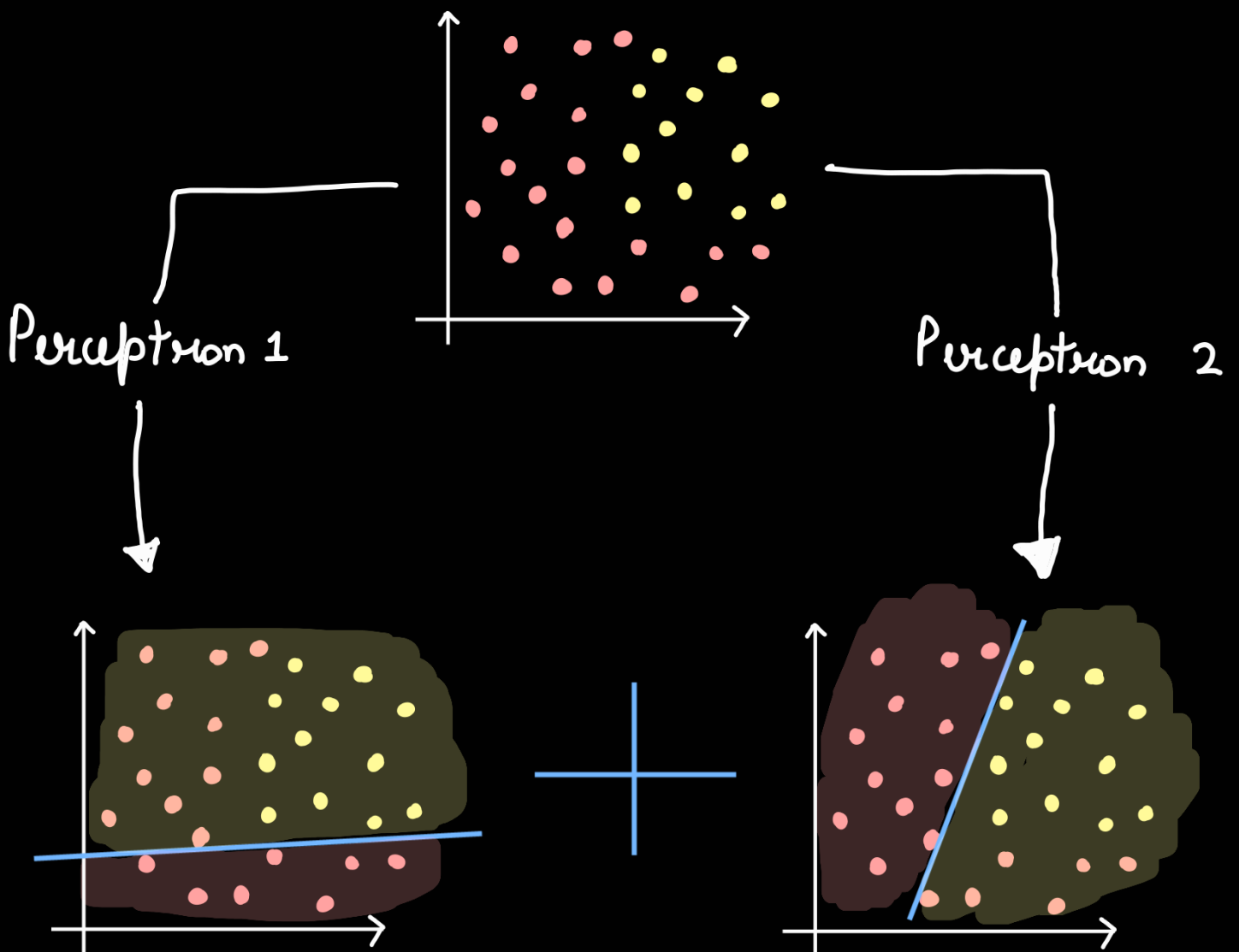
Here the data is not linear, so a line or a linear boundary is not enough to classify the data points



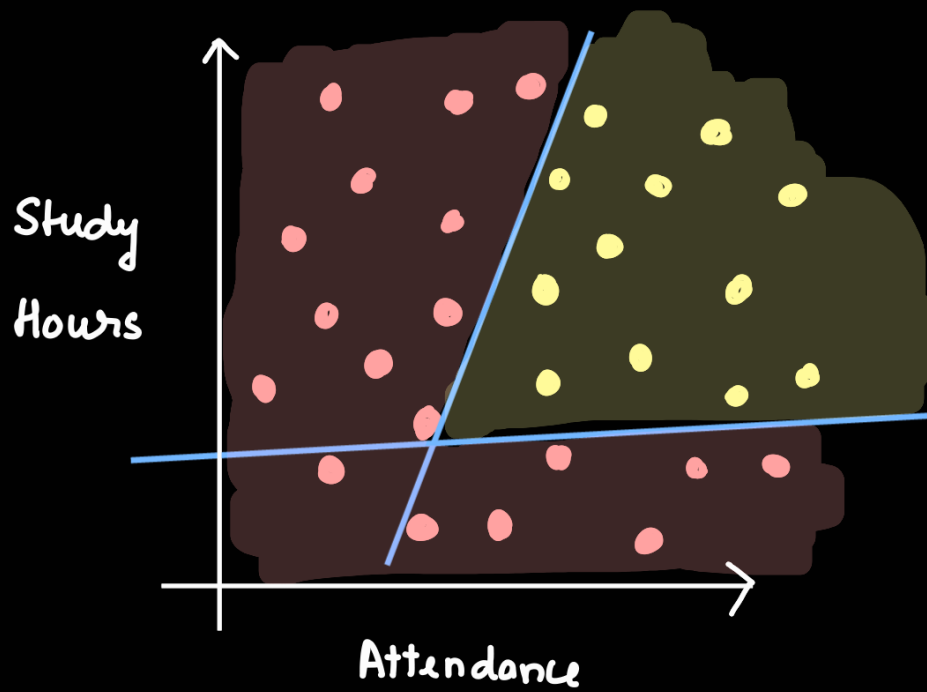
This is how a single perceptron will create the boundary.



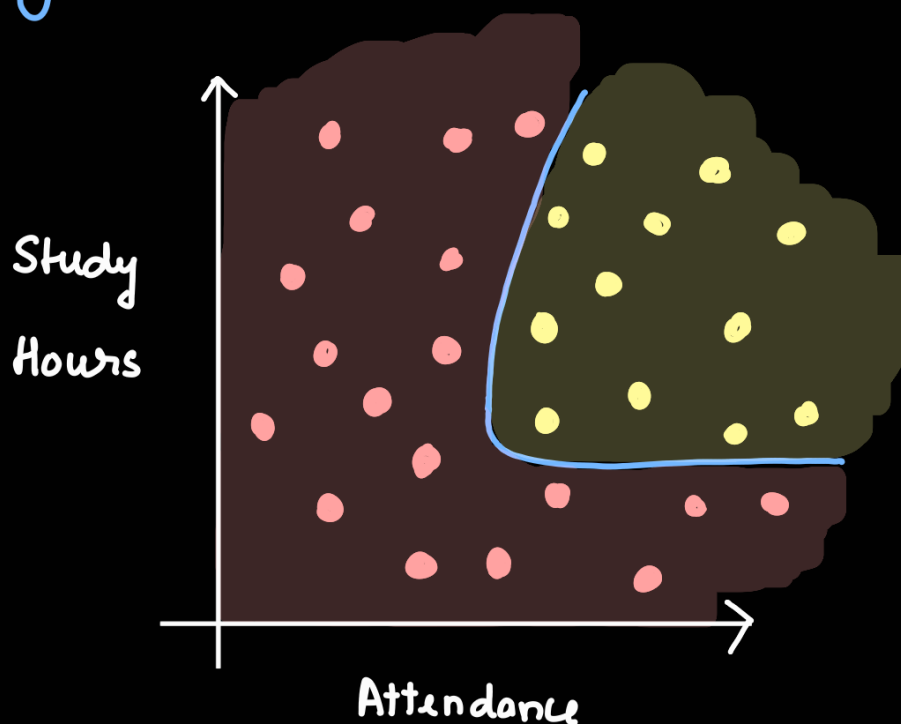
We want this decision boundary.
We can achieve it using MLPs



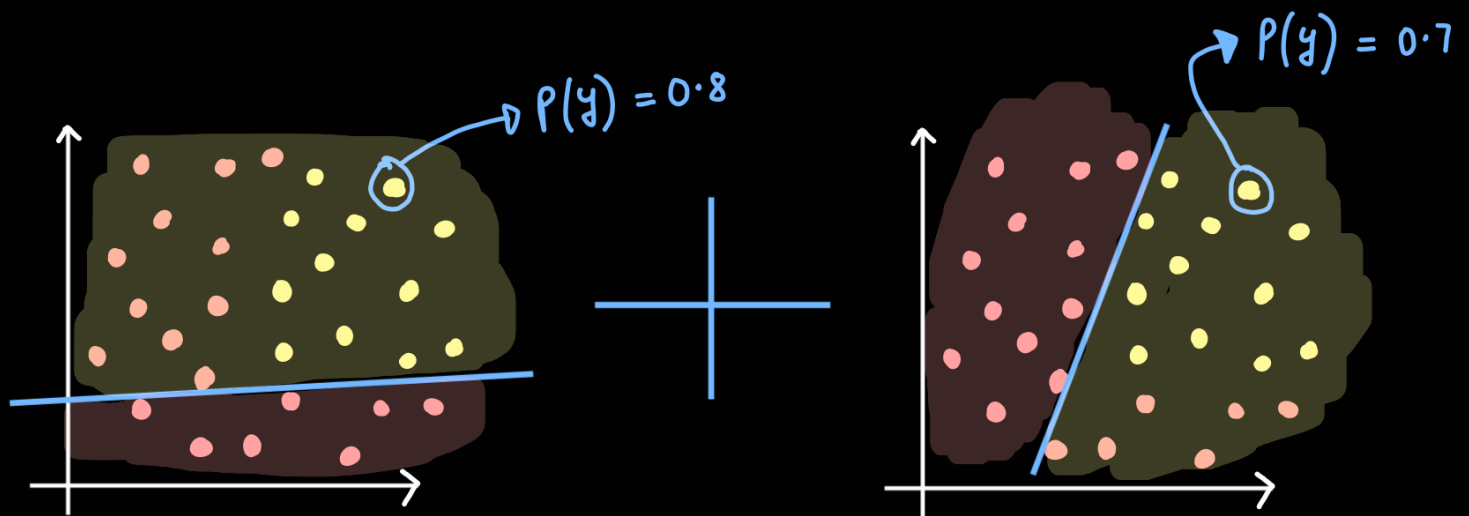
Now lets superimpose both of them on each other.



Now lets smoothen the curve, we will get ÷



Let us examine it mathematically ÷



I randomly selected a student and the $P(y)$ of that student using perceptron 1 is 0.8 and $P(y)$ of that student using perceptron 2 is 0.7.

But how do we combine the probabilities?

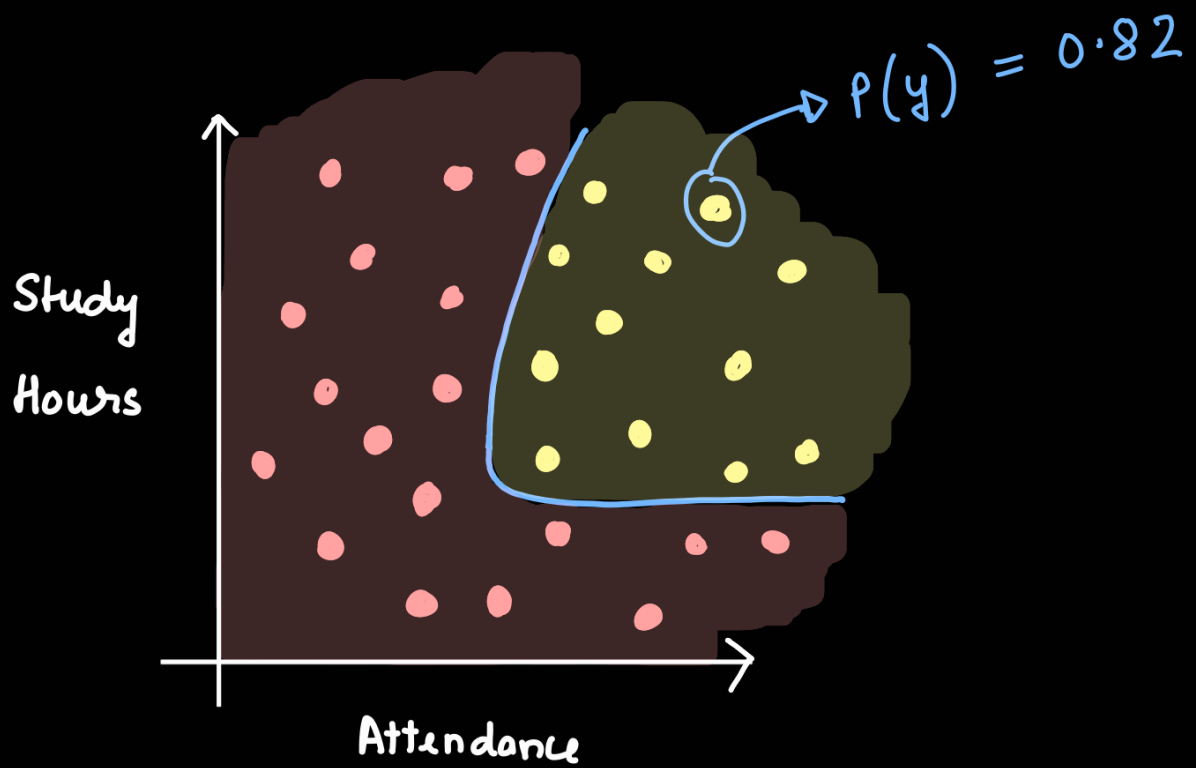
We can't simply just add them because the combined probability will be over 1 [$0.7 + 0.8 = 1.5$].

∴, we also have to apply the sigmoid function after addition

$$\sigma(z) = \frac{1}{1 + e^{-z}} = \frac{1}{1 + e^{-1.5}} = 0.82$$

So, the combined prob. of the student is 0.82

Similarly, find the combined probability for every student to get the new decision boundary.



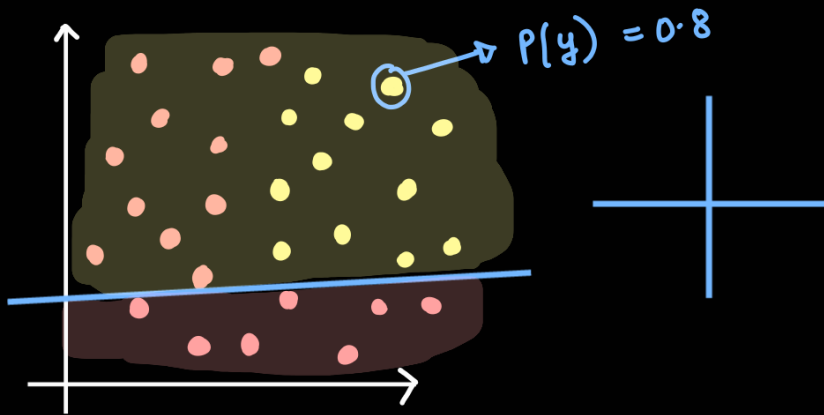
In short, we are finding the linear combination of two perceptrons to find out the new probability

Weights of Perceptron

Suppose that we want to the impact/weightage of Perceptron 1 to be more than that of Perceptron 2

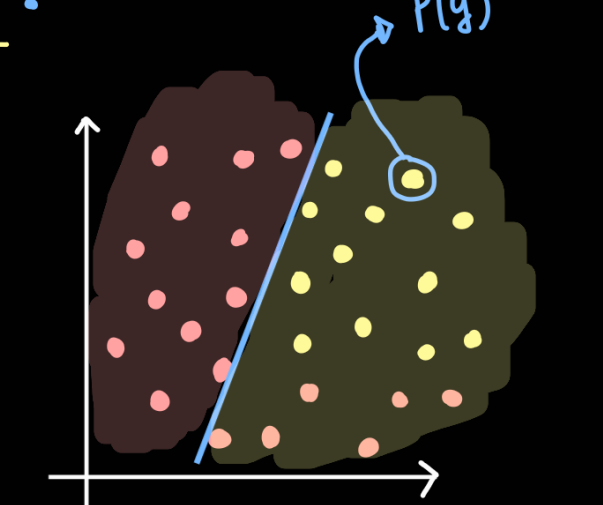
To do so, we can assign weights to the perceptron

Let the weight of Perception 1 be 10 and the weight of Perception 2 be 5.



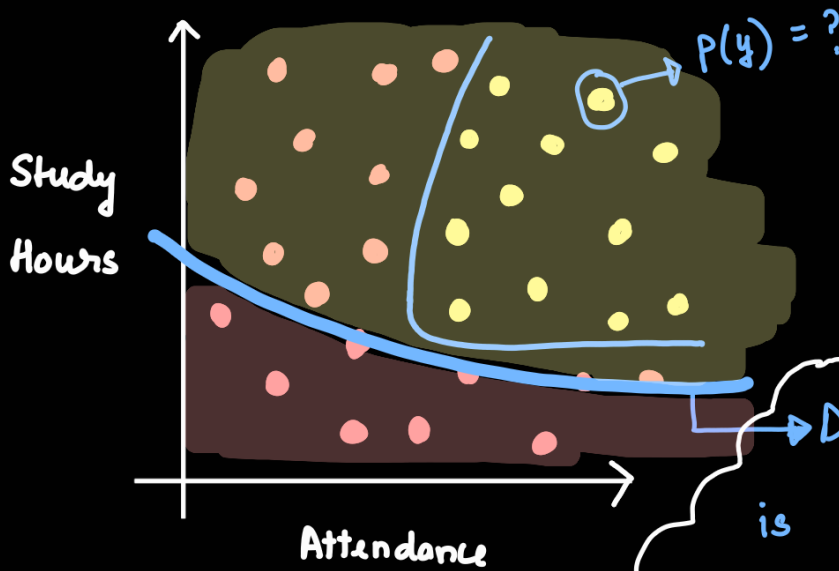
Perception 1

10



5

Perception 2



Decision Boundary is different because Perception 1 have more influence on it as compared to Perception 2

Combined Probability ? [Weighted Addⁿ]

$$0.8(10) + 0.7(5) = 11.5$$

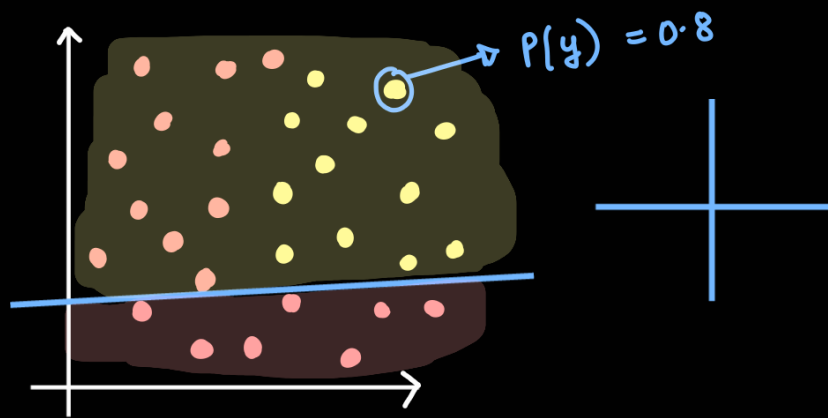
$\sigma(z)$ Sigmoid $f(x^2)$

Will get a value in range 0 - 1

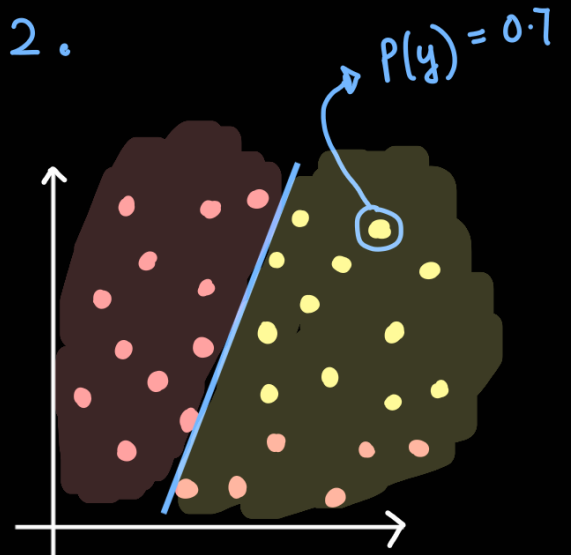
→ This value will be the new combined probability

Bias of Perceptron

What if we also add some bias in the combined result of Perceptron 1 and 2.

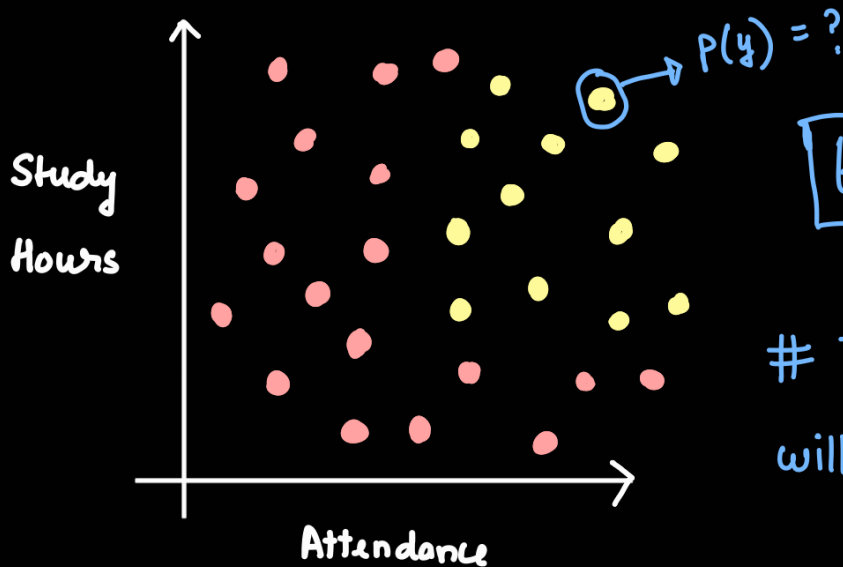


+



Perceptron 1

Perceptron 2



$$\text{bias } (b) = 3$$

The decision boundary will change accordingly.

Combined Probability ?

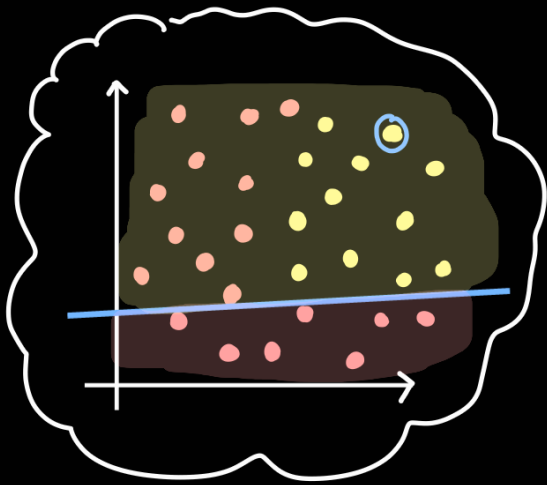
$$Z = 0.8(10) + 0.7(5) + 3$$

Given to sigmoid function

A value ranging
blw 0-1

Combined
Probability

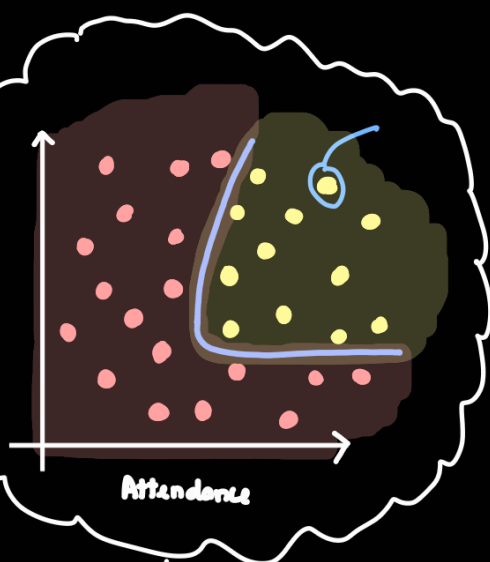
X_1 [Input 1 of P_3]



w_1
10

Study Hours

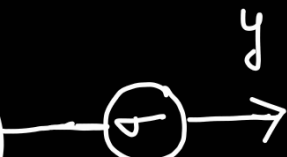
bias = 3



w_2

5

Attendance



y

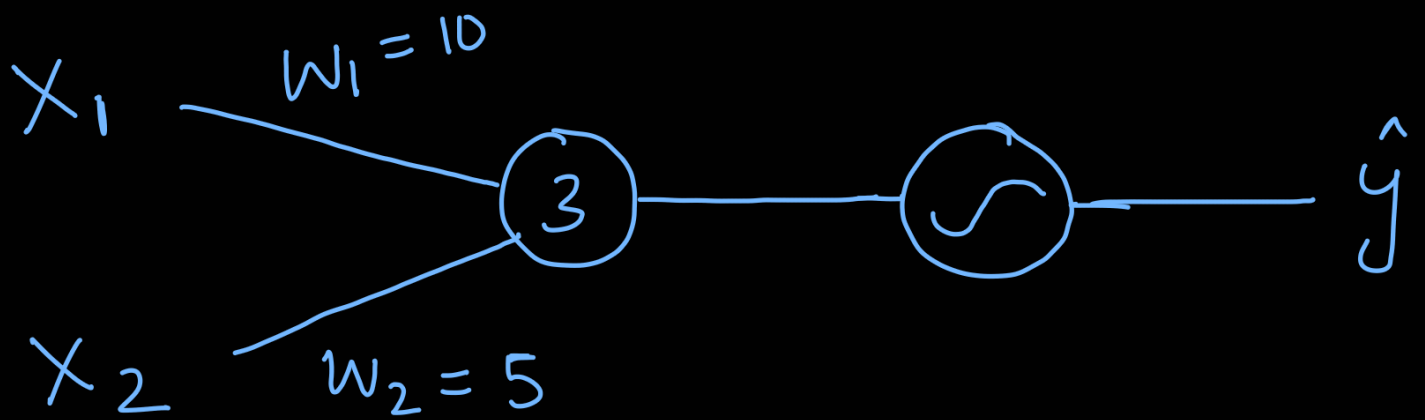
This is actually working like a perceptron.

PERCEPTRON 3 [P3]

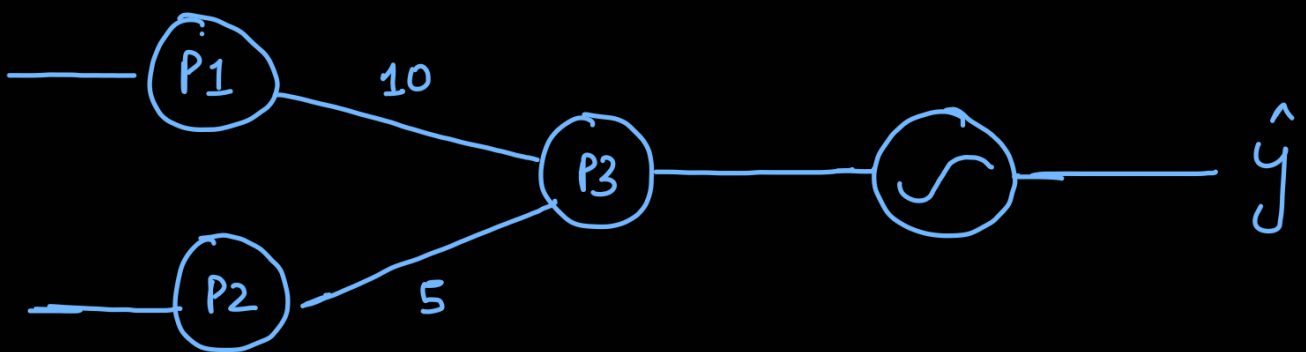
The output of P_1 and P_2 is acting like input for P_3 .

So, we can say that we have a combination of these perceptrons, which we can also call as 'MULTI-LAYER PERCEPTRON'.

We can also draw it as $\div$

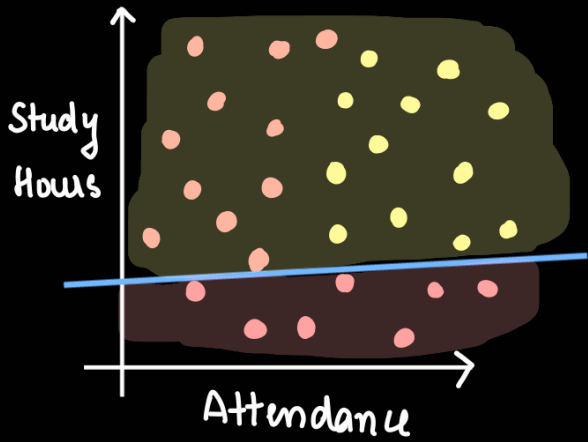


Actually x_1 & x_2 are the output of Perceptron 1 & 2. So, we can also draw it like this $\div$



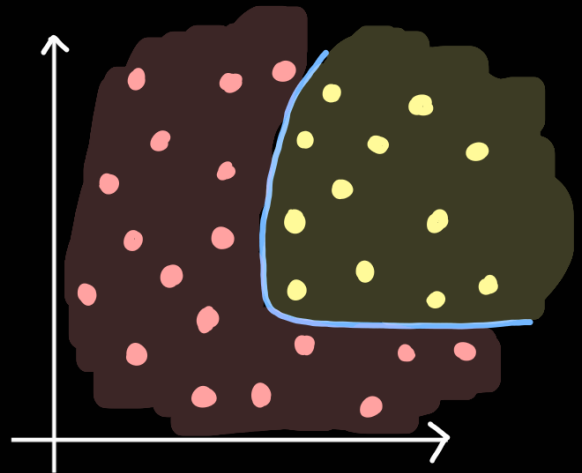
Lets summarise the whole example & see how this is actually working as a MLP :

$$2x + 3y + 6 = 0$$

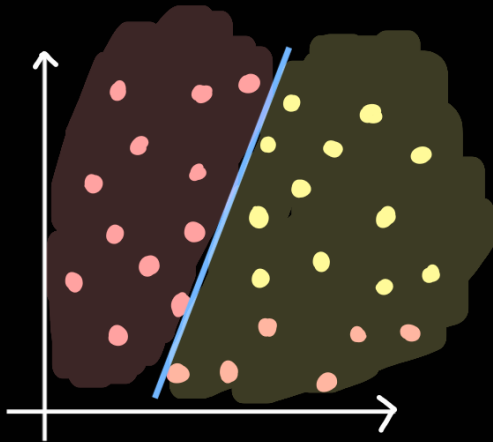


10

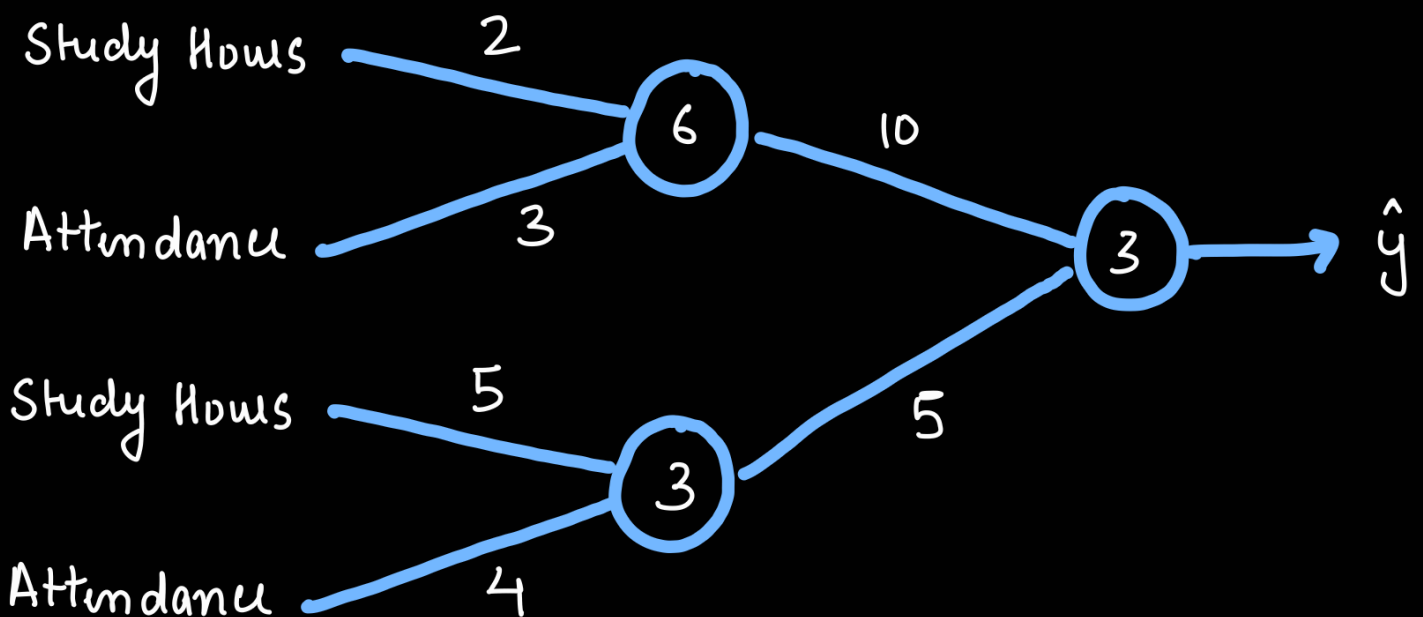
bias = 3



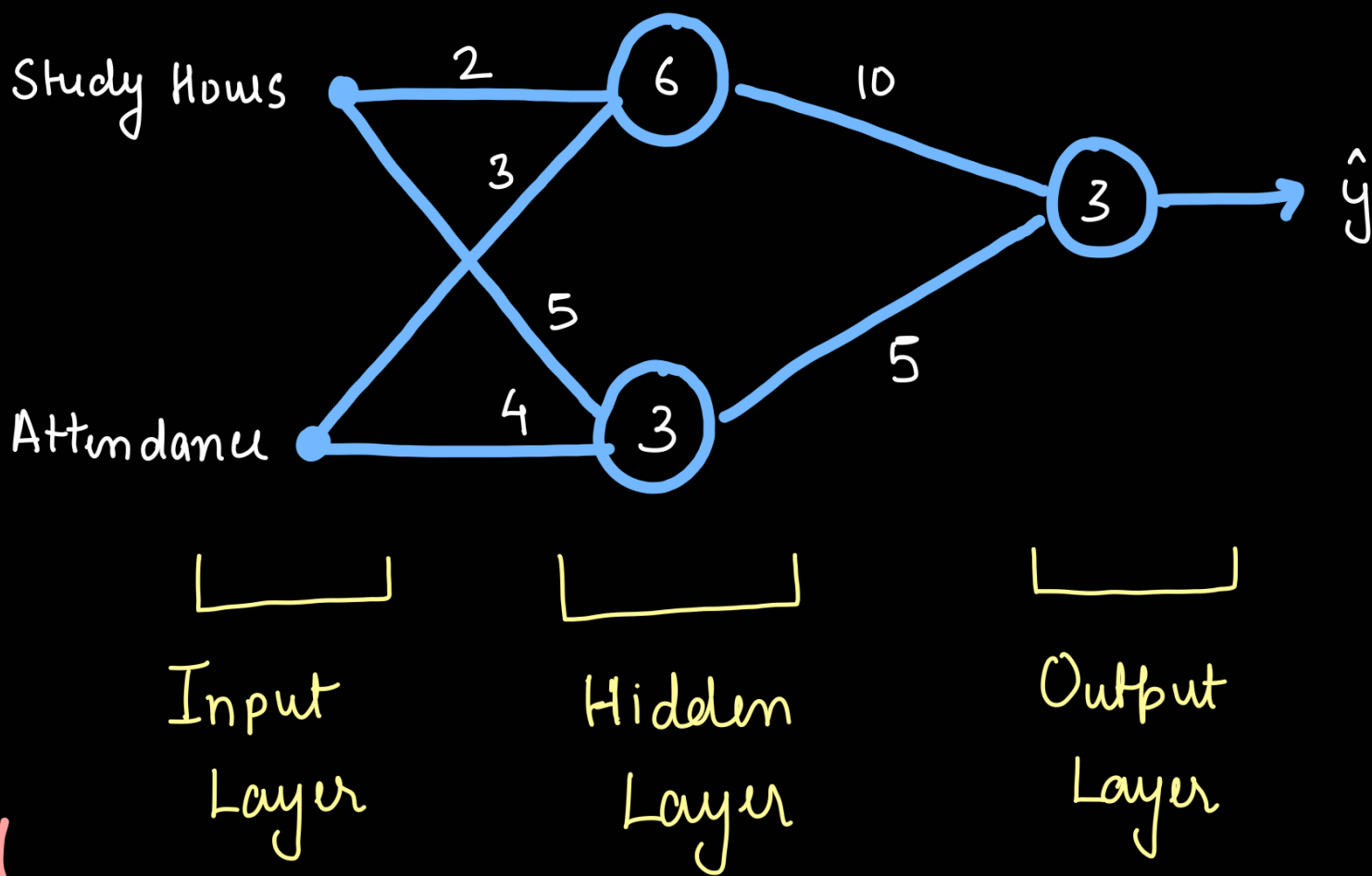
5



$$5x + 4y + 3 = 0$$



We can simplify it more & draw it like this ÷



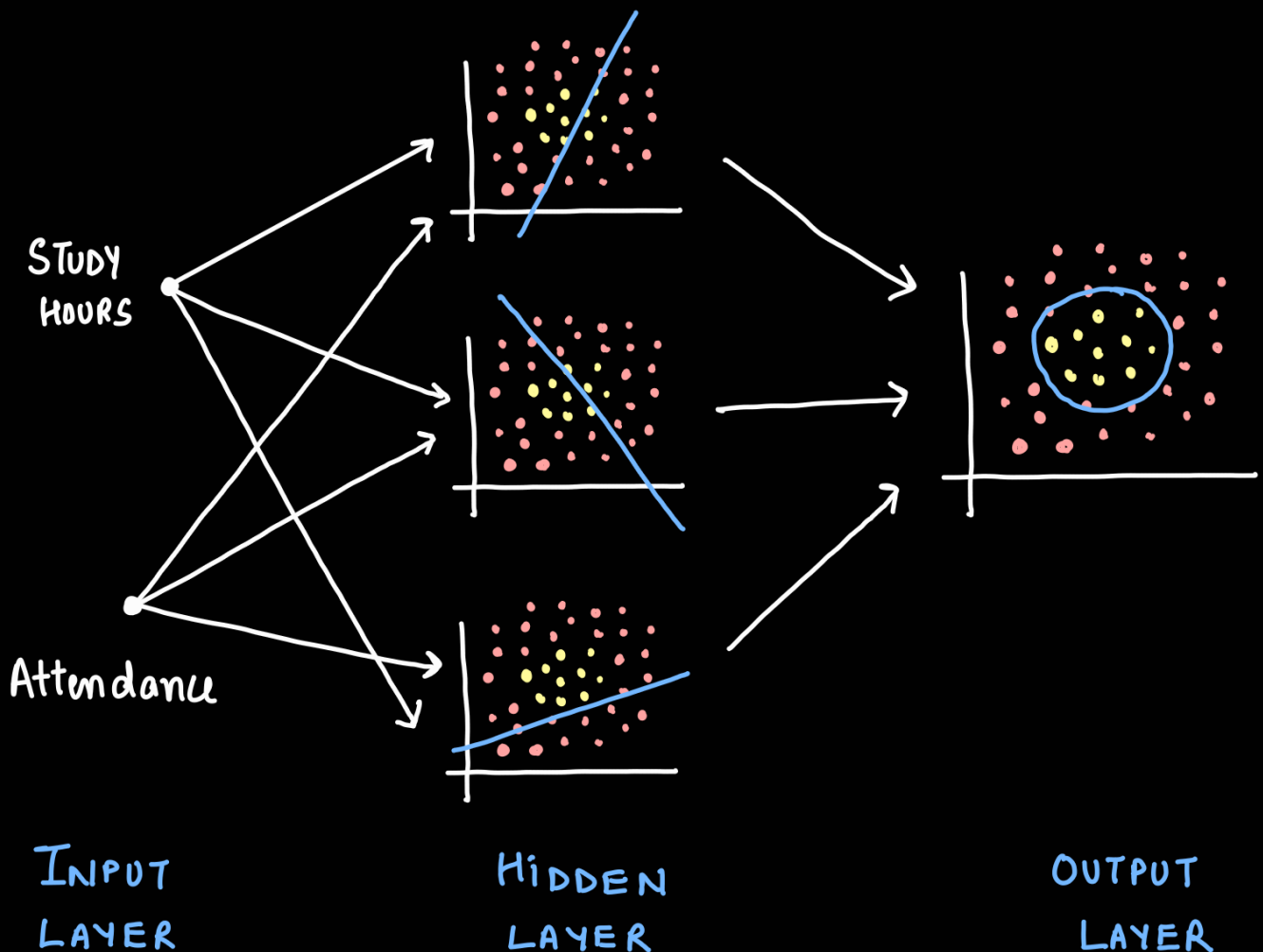
MULTI-LAYER PERCEPTRON

↳ Linear combination of multiple Perceptions.

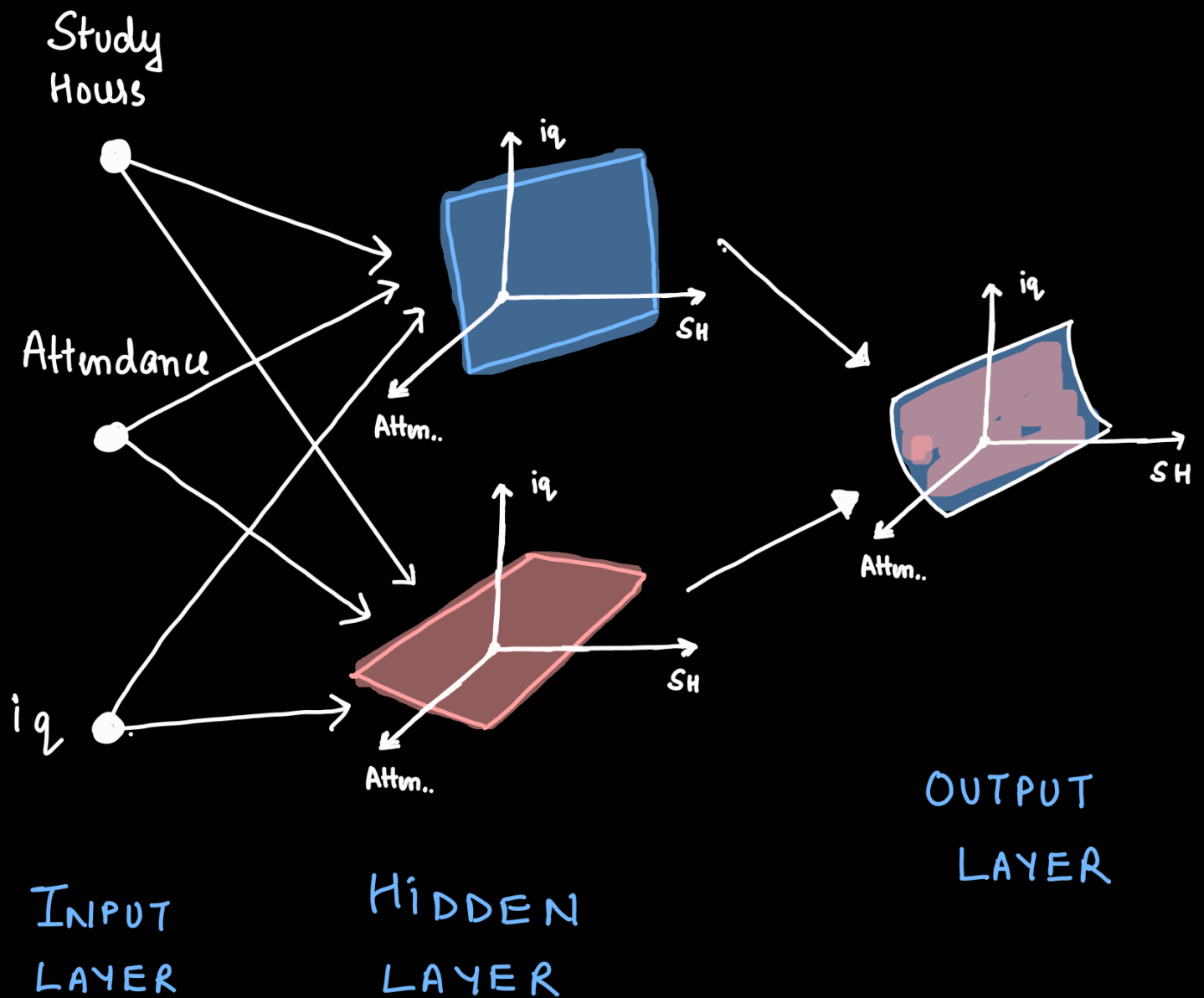
How To CHANGE This ARCHITECTURE

- ① Adding nodes in Hidden Layer.
- ② Adding nodes in input.
- ③ Adding nodes in output layer.
- ④ Add more hidden layers [Deep Neural Network]

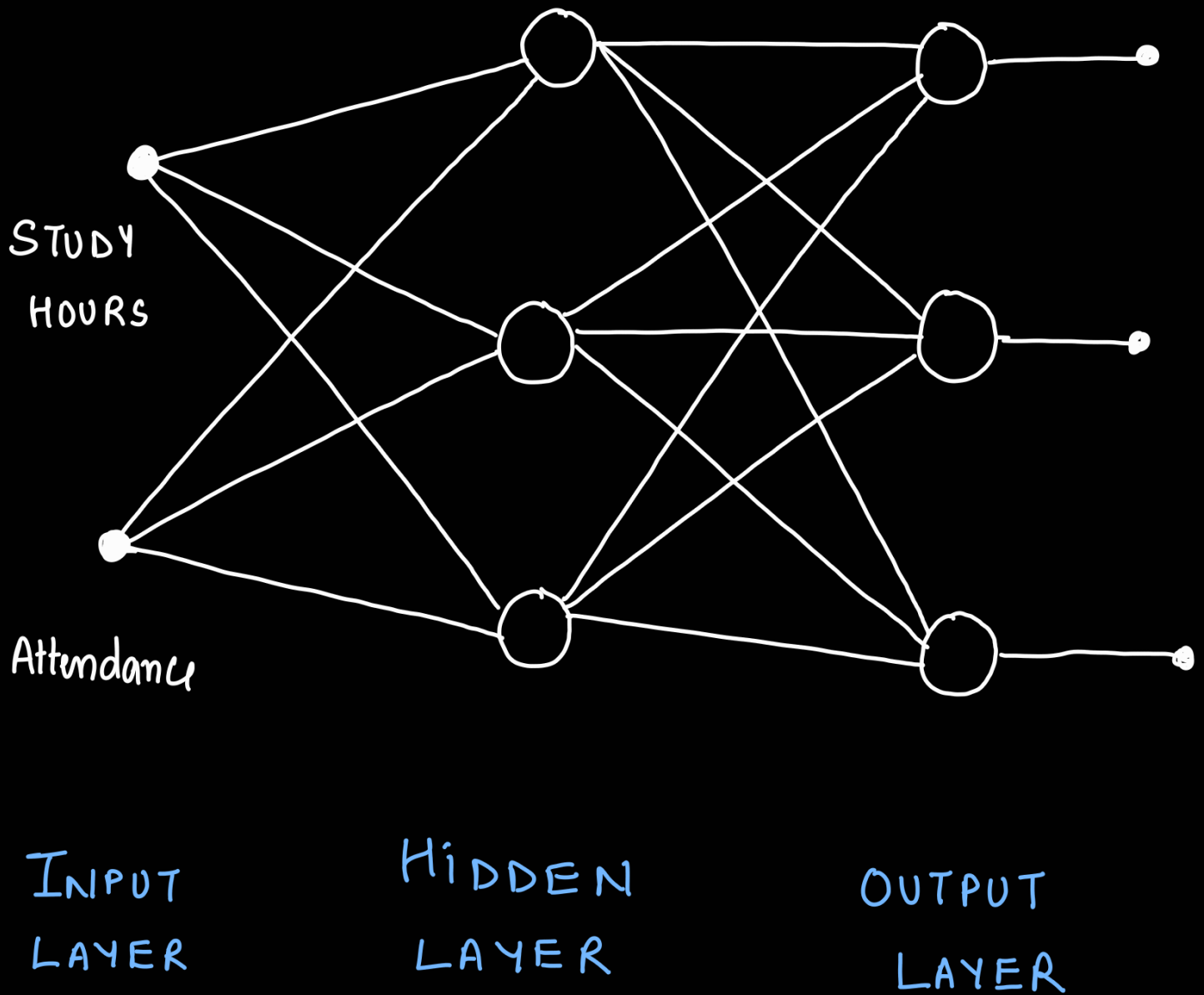
① Adding nodes in Hidden Layer.



② Adding nodes in input.



③ Adding nodes in output layer.



④ Add more hidden layers

